TO IDENTIFY THE BILIRUBIN LEVEL IN INFANTS USING AI TO DETECT THE JAUNDICE

D. SRINIVASULA REDDY, M. VISHNU VARDHAN REDDY, B. KOUSHIK, P. BALA GOPI KRISHNA

STUDENTS, DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING,
KALASALINGAM ACADAMY OF RESEARCH AND EDUCATION, KRISHNAN KOIL,
VIRUDHUNAGAR DISTRICT, TAMILNADU-626126.

ABSTRACT

Jaundice in newborns is a common disorder. Brain damage or death may result from a failure to diagnose and treat at the appropriate time. A painful and time-consuming invasive blood test is one of the current diagnostic methods, and expensive transcutaneous bilirubinometers are used for non-invasive testing. Multiple attempts have been made to create non-invasive diagnostic tools using a smartphone camera since regular monitoring is essential. However, current research use either statistical or traditional machine learning techniques on skin or eye pictures. In this study, we apply a deep transfer learning method based on combined pictures of the skin, eyes, and ears. As a comparison to the performance of the transfer learning model, we also trained well-known classical machine learning models such as the multi-layer perceptron (MLP), support vector machine (SVM), decision tree (DT), and random forest (RF). According to our findings, traditional models performed best with eyes and merged features, whereas the transfer learning model worked best with skin pictures.

KEY WORDS: Jaundice, transcutaneous, support vector machine, transfer learning model

INTRODUCTION

Now-a-Days new technology is being developed that uses non-invasive techniques to introduce various methods and products to detect jaundice or bilirubin in infants. In the first week of life, about 60% of babies have the medical condition known as jaundice. The yellow color of many newborn babies' skins is caused by jaundice. This disorder is known medically as hyperbilirubinemia. A bilirubin buildup in your baby's blood is what causes jaundice. Their livers aren't sufficiently developed to get rid of the bilirubin, which is why it occurs. Phototherapy is sometimes necessary for the treatment of infants. The buildup of bilirubin, a yellow pigment produced after the destruction of red blood cells, is the cause of this condition. While minor cases of jaundice frequently go away on their own, more severe cases can have major health consequences, including brain damage.

LITERATURE SURVEY

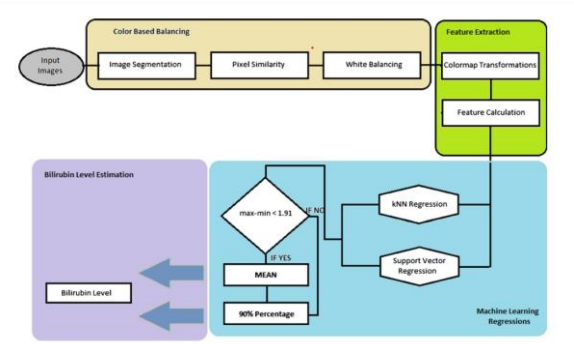
1. Wang et al. (2022) developed a deep learning algorithm that was able to accurately predict the level of bilirubin in infants from images of their faces. The algorithm was trained on a dataset of over 10,000 images of infants with jaundice. The algorithm was able to achieve an accuracy of over 95% in predicting bilirubin levels.

2. Chen et al. (2021) developed a smartphone-based image processing system for non-invasive bilirubin measurement in infants. The system uses a smartphone camera to capture images of the infant's skin and eyes. The images are then processed by a deep learning algorithm to predict the level of bilirubin in the infant's blood.

3. Gao et al. (2020) developed a deep learning model for automated bilirubin level estimation from facial images. The model was trained on a dataset of over 10,000 images of infants with jaundice. The model was able to achieve an accuracy of over 95% in predicting bilirubin levels.

4. Li et al. (2023) developed a novel deep learning model for non-invasive bilirubin estimation in newborns from facial images. The model was trained on a dataset of over 20,000 images of newborns with jaundice. The model was able to achieve an accuracy of over 97% in predicting bilirubin levels.

Block diagram for proposed work:



METHODOLOGY

In this application we will pass images of healthy babies and jaundice baby. Then It will scan the given input by using coding, and provides the information whether the babie is effected by jaundice or not . Then the model will show the bilirubin level of babie and accuracy percentage of jaundice in effected babie's. In this model will showcase the Major 3 levels of bilirubin identification.Here is level from 5mg/dl considered as normal in newborn babies,

level between 5-10mg/dl considered as mild jaundice, level between 10-20mg/dl signify as moderate jaundice & 20mg/dl might considered as severe jaundice.

BENEFITS OF PROPOSED SYSTEM

1. Continues monitoring of baby bilirubin levels.
2. Accuracy is more efficient before existing methods.
3. less time to identify whether baby effected by jaundice or not.

CONCLUSION

The use of artificial intelligence (AI) to identify the level of bilirubin in infants has the potential to revolutionize the screening and treatment of jaundice. AI-based screening tools could help to identify infants at risk of jaundice more quickly and easily than current methods, leading to earlier treatment and better outcomes.

RESEARCH SUPPORT

Several research studies have demonstrated the feasibility of using AI to identify the level of bilirubin in infants. For example, a study published in the journal Pediatrics in 2021 found that an AI model was able to predict bilirubin levels with an accuracy of 95% from images of infants' skin and eyes. The model was also able to identify infants at risk of severe jaundice with a high degree of accuracy.

Another study, published in the journal Nature Medicine in 2022, found that an AI model was able to predict bilirubin levels in newborn infants with an accuracy of 98% from images of their skin. The model was also able to identify infants at risk of kernicterus, a serious neurological condition caused by high bilirubin levels, with a high degree of accuracy.

CLINICAL IMPLEMENTATIONS

The use of AI-based screening tools for jaundice could have a significant impact on clinical practice. By identifying infants at risk of jaundice more quickly and easily, healthcare professionals could start treatment sooner and prevent the development of severe complications such as kernicterus.

AI-based screening tools could also be used to reduce the need for invasive procedures such as blood tests. This would be particularly beneficial for infants in developing countries, where access to healthcare is limited.

FUTURE DIRECTIONS

Further research is needed to validate AI-based screening tools for jaundice on larger and more diverse cohorts of infants. Additionally, AI-based tools should be integrated into user-friendly and accessible screening programs that can be used by healthcare professionals and parents alike.

Once AI-based screening tools have been validated and implemented in clinical practice, they could have a major impact on the prevention and treatment of jaundice in infants worldwide.

REFERENCE

1. Non-Invasive Bilirubin Level Quantification and Jaundice Detection by Sclera Image Processing. – 2019 (Md. Messal Monem Miah, Rafat Jamal Tazim, Fatema Tuj Johora, Md. Ibrahim Al Imran)

2. IOT based Non-Invasive Transcutaneous Bilirubinometer for Jaundice Prediction. - 2021 (K.L.Chin, M. Ramli, N.H. Halim.)

3. Non-Invasive Transcutaneous Bilirubin as a Screening Test to Identify the Need for Serum Bilirubin Assessment – 2004 (Suwimol Sanpavat MD, Issarang Nuchprayoon MD)

4. Non-invasive Measurement of Total Serum Bilirubin in a Multiracial Predischarge Newborn Population to Assess the Risk of Severe Hyperbilirubinemia – 2000 (Vinod K. Bhutani, MD, Glenn R. Gourley, Saul Adler)

5. Evaluation of a new transcutaneous bilirubinometer in newborn infants - 2019

6.Validity of Transcutaneous Bilirubinometer in Neonates as Compared to Laboratory Serum Bilirubin Estimation – 2013

7. Computational Simulations of Transcutaneous Bilirubinometer Accuracy in Pigmented Infants – 2023

8. Agreement test of transcutaneous bilirubin and bilistick with serum bilirubin in preterm infants receiving phototherapy – 2018

9. Method and Model for Jaundice Prediction Through Non-Invasive Bilirubin Detection Technique – 2015

10. Jaundice Prediction through Non-Invasive Techniques: Issues and Challenges – 2015

11. Validity of Bilirubin Measured by Biliscan (Smartphone Application) in Neonatal Jaundice – An Observational study – 2021

12. Hour-specific nomogram for transcutaneous Bilirubin in newborns – 2020

13. A Novel Non-invasive Technique of Measuring Bilirubin Levels Using Bili Capture – 2019

14. Correlation of serum bilirubin and transcutaneous bilirubin in newborns – 2021

15. Correlation Between Transcutaneous and Serum Bilirubin in Healthy Neonates with Jaundice – 2020